Highlights

- Open, checksummed model of 5025 graded breast cancers from 7 public cohorts
- Logistic regression matched gradient boosting for grade, internally and externally
- External calibration failed in opposite directions; an intercept update fixed it
- Leakage shortcuts inflated macro-AUC by up to +0.297; MRI radiomics added nothing
- Saudi case-mix implies more grade 3; Western models over-predict it locally